VELAMMAL KNOWLEDGE PARK | BODHI IIT/NEET ACADEMY

XII BIOLOGY

IMPORTANT TOPICS REVISION

SESSION 2025 – 26 | CBSE BOARD EXAM

Ch. 1 Sexual Repro. in Flowering Plants

Ch. 2 Human Reproduction

Ch. 3 Reproductive Health

Ch. 4 Principles of Inheritance & Variation

Ch. 5 Molecular Basis of Inheritance

Ch. 6 Evolution

Ch. 7 Human Health and Diseases

Ch. 8 Microbes in Human Welfare

Ch. 9 Biotechnology: Principles & Processes

Ch. 10 Biotechnology & Its Applications

Ch. 11 Organisms and Populations

Ch. 12 Ecosystem

Ch. 13 Biodiversity and Conservation

★★★	Very Important	Always in board exam
★★	Important	Frequently asked
★	Good to Know	Concept clarity / MCQs

TABLE OF CONTENTS

No.	Chapter Name	Topics	Day
1	Sexual Reproduction in Flowering Plants	5	Day 1
2	Human Reproduction	7	Day 2
3	Reproductive Health	3	Day 4
4	Principles of Inheritance and Variation	7	Day 2
5	Molecular Basis of Inheritance	7	Day 3
6	Evolution	5	Day 3
7	Human Health and Diseases	5	Day 4
8	Microbes in Human Welfare	6	Day 4
9	Biotechnology: Principles and Processes	3	Day 5
10	Biotechnology and Its Applications	4	Day 5
11	Organisms and Populations	3	Day 6
12	Ecosystem	4	Day 6
13	Biodiversity and Conservation	6	Day 6

QUICK SUMMARY — KEY CONCEPTS AT A GLANCE

Chapter	Core Concept
Ch.1 Sexual Repro.	Pollen → Pollen tube → Double Fertilisation (syngamy + triple fusion) → Seed & Fruit. Apomixis = seed without fertilisation.
Ch.2 Human Repro.	Spermatogenesis (testes) & Oogenesis (ovaries). Fertilisation in ampulla → Blastocyst implants in endometrium ∼7 days post-fertilisation.
Ch.3 Repro. Health	MTP legal up to 20 weeks. Barrier, hormonal, IUD, surgical contraception. STDs: gonorrhoea, syphilis, HIV. ART for infertility.
Ch.4 Genetics	Mendel's Laws: Segregation & Independent Assortment. Codominance in ABO. XY (humans), ZW (birds) sex determination.
Ch.5 Mol. Basis	DNA double helix (Watson & Crick). Central Dogma: DNA→RNA→Protein. Lac operon = inducible gene regulation.
Ch.6 Evolution	Miller–Urey proved chemical evolution. Darwin's Natural Selection. Hardy–Weinberg: no evolution in equilibrium population.
Ch.7 Health & Disease	Innate vs Adaptive immunity. B-cells→antibodies (humoral); T-cells = CMI. HIV destroys $CD4^+$ T-helper lymphocytes.
Ch.8 Microbes	Lactobacillus (curd), Saccharomyces (bread/alcohol), Penicillium (penicillin), Methanogens (biogas), Rhizobium (N-fixation).
Ch.9 Biotech Prin.	RE cuts DNA, Ligase joins, Vectors carry rDNA. PCR amplifies DNA (Taq polymerase). pBR322 = first recombinant plasmid.
Ch.10 Biotech App.	Bt cotton (cry genes), Golden Rice (Vit A), Humulin (insulin from *E. coli*). GEAC approves GMOs in India.
Ch.11 Organisms	Exponential: $\frac{dN}{dt} = rN$. Logistic (S-curve) with K. Gause's competitive exclusion. Mutualism, Predation, Parasitism.
Ch.12 Ecosystem	NPP = GPP − R. 10% Energy Law (Lindeman). Pyramid of energy always upright. Carbon & Nitrogen biogeochemical cycles.
Ch.13 Biodiversity	India: 4 hotspots. In-situ (national parks) vs Ex-situ (seed banks, zoos). Evil Quartet = 4 causes of biodiversity loss.

Chapter 1 | Sexual Reproduction in Flowering Plants

Day 1 · 5 Topics

#1 Flower — A Fascinating Organ of Angiosperms

★★★ Very Imp.

★★★ **Structure: Sepals (calyx), Petals (corolla), Androecium (male), Gynoecium (female)**

★★★ **Androecium: Stamens = Filament + Anther (contains 4 microsporangia)**

★★★ **Gynoecium: Pistil = Stigma + Style + Ovary (contains megasporangia)**

★★★ **Bisexual: both whorls present; Unisexual: only one whorl (monoecious / dioecious plants)**

★★★ **Placentation types: Marginal, Axile, Parietal, Basal, Free central, Superficial**

- Actinomorphic = radial symmetry; Zygomorphic = bilateral symmetry (e.g., pea flower)
- Epigynous (inferior ovary), Hypogynous (superior ovary), Perigynous (half inferior)

#2 Pre-Fertilisation: Structures and Events

★★★ Very Imp.

★★★ **Microsporogenesis: Pollen Mother Cell (PMC, 2n) $\xrightarrow{\text{meiosis}}$ 4 haploid microspores (microspore tetrad)**

★★★ **Pollen grain wall: Exine (outer; made of sporopollenin — most resistant biological material) + Intine (inner; cellulose/pectin)**

★★★ **Mature pollen = 2-celled (vegetative + generative) OR 3-celled (vegetative + 2 male gametes after mitosis)**

★★★ **Megasporogenesis: MMC (2n) $\xrightarrow{\text{meiosis}}$ 4 megaspores; 3 degenerate, 1 functional (chalazal end)**

★★★ **Female Gametophyte (Embryo sac): 7-celled, 8-nucleated**

- 3 antipodal cells (antipodal end) + 2 synergids + 1 egg cell (micropylar end) + 1 central cell (2 polar nuclei)

★★★ **Pollination types: Autogamy (same flower), Geitonogamy (different flower, same plant), Xenogamy (different plant)**

★★ Agents: Anemophily (wind), Hydrophily (water), Entomophily (insects), Ornithophily (birds), Chiropterophily (bats)

★★ Outbreeding devices: Self-incompatibility, Dioecy, Dichogamy (different maturity times), Heterostyly

> ***Note:*** *Sporopollenin is so resistant it survives strong acids, alkalis, and high temperatures.*

#3 Double Fertilisation

★★★ Very Imp.

★★★ **Pollen tube pathway: Stigma → Style → Ovary → Micropyle → Embryo sac (enters via filiform apparatus of synergid)**

★★★ **Syngamy: 1st male gamete (n) + Egg cell (n) → Zygote (2n)**

★★★ **Triple fusion: 2nd male gamete (n) + 2 Polar nuclei (n+n) → Primary Endosperm Nucleus — PEN (3n)**

★★★ **Double Fertilisation = Syngamy + Triple Fusion — UNIQUE TO ANGIOSPERMS (discovered by Nawaschin, 1898)**

★★ Chalazogamy = pollen tube enters through chalaza (not micropyle). Porogamy = through micropyle (usual)

#4 Post-Fertilisation: Structures and Events

★★★ Very Imp.

★★★ **Zygote (2n) → Embryo; PEN (3n) → Endosperm — endosperm develops BEFORE embryo**

★★★ **Endosperm types: Nuclear (free nuclei first — e.g., coconut water = liquid endosperm), Cellular, Helobial**

★★★ **Dicot embryo: Cotyledons (2), Radicle, Plumule, Epicotyl, Hypocotyl**

★★★ **Monocot embryo: Scutellum (1 cotyledon), Coleoptile (encloses plumule), Coleorhiza (encloses radicle)**

★★★ **Ovule → Seed; Ovary → Fruit; Ovary wall → Pericarp**

- Parthenocarpy: Fruit formation without fertilisation (seedless fruits) — induced by auxin (e.g., banana, grapes)

#5 Apomixis and Polyembryony

★★ Important

★★ Apomixis: Seed production WITHOUT fertilisation (from diploid egg cell or nucellus without meiosis)

- Significance: Fixes hybrid vigour, avoids meiosis; common in Asteraceae, grasses

★★ Polyembryony: More than one embryo per seed (e.g., Citrus, Mango, Groundnut)

> ***Note:*** *Nucellar polyembryony in Citrus: extra embryos are exact clones of the mother plant — important in agriculture.*

Chapter 2 | Human Reproduction

Day 2 · 7 Topics

#1 The Male Reproductive System

★★★ Very Imp.

★★★ **Testes in scrotum at 2–2.5°C below body temp — essential for spermatogenesis**

★★★ **Sperm pathway: Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra**

★★★ **Accessory glands: Seminal vesicles (60% of semen volume), Prostate (30%), Bulbourethral/Cowper's glands**

★★ Semen = spermatozoa + secretions; normal ejaculate contains 200–300 million sperms in 2–5 mL

#2 The Female Reproductive System

★★★ Very Imp.

★★★ **Ovaries (primary organs): produce ova + oestrogen + progesterone**

★★★ **Fallopian tube: Infundibulum (fimbriae) → Ampulla (site of fertilisation) → Isthmus → Uterus**

★★★ **Uterine wall layers: Perimetrium (outer) → Myometrium (thick muscle) → Endometrium (inner; shed in menstruation)**

- Cervix = lower narrow part of uterus; cervical canal connects to vagina

#3 Gametogenesis

★★★ Very Imp.

★★★ **Spermatogenesis: Spermatogonia (2n) → Primary spermatocyte → Secondary spermatocyte → Spermatids $\xrightarrow{\text{spermiogenesis}}$ Spermatozoa**

★★ Sertoli cells: nourish developing sperms; **Leydig (interstitial) cells**: secrete testosterone

★★★ **Sperm structure: Head (acrosome with hydrolytic enzymes + nucleus) + Middle piece (mitochondrial spiral — energy) + Tail (flagellum)**

★★★ **Oogenesis: Oogonium → Primary oocyte (arrested at Prophase-I at birth) $\xrightarrow{\text{puberty, LH surge}}$ Secondary oocyte → Ovum**

★★ Oogenesis produces 1 functional ovum + 3 non-functional polar bodies (unequal cytokinesis)

★★★ **Folliculogenesis: Primordial → Primary → Secondary → Graafian follicle $\xrightarrow{\text{ovulation}}$ Corpus luteum → Corpus albicans**

#4 Menstrual Cycle

★★★ Very Imp.

★★★ **Duration: ~28 days; Menarche = first menstruation (puberty ~13 yrs); Menopause ~50 years**

★★★ **Phase 1 — Menstrual (Day 1–5): Endometrium shed; low oestrogen + progesterone**

★★★ **Phase 2 — Follicular/Proliferative (Day 6–13): FSH rises → follicle grows → oestrogen rises → endometrium thickens**

★★★ **Phase 3 — Ovulatory (Day 14): LH surge triggers ovulation (secondary oocyte + 1st polar body released)**

★★★ **Phase 4 — Luteal/Secretory (Day 15–28): Corpus luteum secretes progesterone → endometrium maintained; if no fertilisation → corpus luteum degenerates → menstruation**

Note: *Cyclic menstruation is absent during pregnancy (hCG maintains corpus luteum) and during lactation (lactational amenorrhoea).*

#5 Fertilisation and Implantation

★★★ Very Imp.

★★★ **Site of fertilisation: Ampulla of fallopian tube**

★★★ **Capacitation: Sperm acquire fertilising ability in female reproductive tract**

★★★ **Acrosomal reaction: Acrosome releases hyaluronidase + acrosin → zona pellucida dissolved → sperm fuses with egg plasma membrane**

★★★ **Cortical reaction: Ca^{2+} release → zona pellucida hardens (zona reaction) → blocks polyspermy**

★★★ **Cleavage: Zygote → 2-cell → 4-cell → 8-cell → Morula (16–32 cells, solid) → Blastocyst (hollow; inner cell mass + trophoblast)**

★★★ **Implantation: Blastocyst embeds in endometrium ∼7 days post-fertilisation; trophoblast secretes hCG**

#6 Pregnancy and Embryonic Development

★★★ Very Imp.

★★★ **Placenta: Formed by chorion (foetal) + decidua basalis (maternal endometrium); fully formed by ∼3rd month**

★★★ **Placental functions: Exchange of nutrients, O_2, CO_2, waste; acts as endocrine organ (hCG, hPL, oestrogen, progesterone)**

★★ hCG (human chorionic gonadotropin): detected in urine/blood = basis of pregnancy test; peak at 10–12 weeks

- Three germ layers: Ectoderm (skin, nervous system), Mesoderm (muscles, skeleton, heart), Endoderm (gut, lungs)

★★★ **Key milestones: Heartbeat at 4 weeks; Limb buds at 6 weeks; Movements felt at ∼5 months; Full term = 280 days (40 weeks)**

#7 Parturition and Lactation

★★★ Very Imp.

★★★ **Parturition (childbirth): Foetal signals → prostaglandins + oxytocin from posterior pituitary → uterine contractions (positive feedback)**

★★★ **Colostrum: First yellowish milk; rich in IgA antibodies → passive immunity to newborn; also contains proteins and Vit A**

★★ Prolactin (anterior pituitary): stimulates milk **production**; Oxytocin: stimulates milk **ejection** (let-down reflex)

Note: *Lactational amenorrhoea: High prolactin suppresses GnRH → no ovulation for ∼6 months post-delivery (natural contraception).*

Chapter 3 | Reproductive Health

Day 4 · 3 Topics

#1 Problems and Strategies

★★ Important

★★ WHO definition: Total well-being in all aspects of reproduction — physical, emotional, behavioural, social

- National programs: Family Planning (1952 — first in world), RNTCP (TB), NACP (AIDS)

★★★ **Amniocentesis: Foetal cells from amniotic fluid → detect chromosomal abnormalities (Down syndrome) and metabolic disorders; BANNED for sex determination in India (Pre-Natal Diagnostic Techniques Act)**

#2 Population Explosion, Birth Control, MTP

★★★ Very Imp.

★★★ **India: >1.4 billion people (2nd most populous); doubling time was ~33 years in 1960s**

★★★ **Contraceptive categories:**

- **Natural**: Periodic abstinence (rhythm method), Coitus interruptus, Lactational amenorrhoea
- **Barrier**: Male condom (also STD protection), Female condom, Diaphragm, Cervical cap, Vault
- **IUDs**: Non-medicated (Lippes loop); Copper IUDs (Cu-T, Cu7, Multiload 375); Hormonal (LNG-20 — releases levonorgestrel)
- **Oral pills**: Combined (oestrogen + progesterone) — inhibit ovulation; Progestin-only (POP); Emergency (within 72 hrs)
- **Injectables/Implants**: Depo-Provera; Norplant (subdermal implant)
- **Surgical (permanent)**: Vasectomy (male — vas deferens cut/tied); Tubectomy (female — fallopian tube cut/tied/sealed)

★★★ **MTP (Medical Termination of Pregnancy): Legal in India up to 20 weeks (MTP Act 1971, amended 2021); requires medical opinion**

★★ Ideal contraceptive: No/minimal side effects, effective, reversible, easily available, user-friendly, no coital interference

#3 STDs and Infertility

★★★ Very Imp.

★★★ **STDs (Sexually Transmitted Diseases):**

- Bacterial: Gonorrhoea (*Neisseria gonorrhoeae*), Syphilis (*Treponema pallidum*), Chlamydiasis (*Chlamydia trachomatis*)
- Viral: HIV/AIDS, Genital herpes (HSV-2), Genital warts (HPV — cervical cancer risk), Hepatitis-B
- Others: Trichomoniasis (*Trichomonas vaginalis* — protozoan), Candidiasis (fungal — *Candida albicans*)

★★★ **EXCEPT hepatitis-B, genital herpes, and HIV — all STDs are curable if detected early and treated**

★★ Infertility: Inability to conceive after 12 months of regular unprotected intercourse

★★★ **ART (Assisted Reproductive Technologies):**

- IVF-ET: Fertilisation in lab → embryo transfer to uterus ('test-tube baby'; Louise Brown, 1978)
- ZIFT: Zygote Intra Fallopian Transfer; GIFT: Gamete Intra Fallopian Transfer
- ICSI: Intra Cytoplasmic Sperm Injection (single sperm injected into egg)
- IUI: Intra Uterine Insemination (donor/husband's processed sperm injected into uterus)

Chapter 4 | Principles of Inheritance and Variation

Day 2 · 7 Topics

#1 Mendel's Laws of Inheritance

★★★ Very Imp.

★★★ **Mendel chose 7 contrasting traits in *Pisum sativum* (garden pea) — all happened to be on different chromosomes**

★★★ **Law of Dominance: In a heterozygote, dominant allele is expressed; recessive allele is masked**

★★★ **Law of Segregation (Purity of Gametes): Alleles separate during gamete formation; each gamete carries only one allele per locus**

★★★ **Law of Independent Assortment: Genes on non-homologous chromosomes assort independently during meiosis**

#2 Inheritance of One Gene

★★★ Very Imp.

★★★ **Monohybrid cross: Tt × Tt → Genotypic 1TT:2Tt:1tt; Phenotypic 3:1**

★★★ **Test cross: Dominant phenotype × homozygous recessive (tt) → ratio 1:1 = heterozygous; ratio 1:0 = homozygous dominant**

★★★ **Incomplete dominance: F_1 intermediate phenotype (e.g., Antirrhinum: RR red × rr white → Rr pink); F_2 = 1 red : 2 pink : 1 white**

★★★ **Codominance: Both alleles expressed equally (e.g., ABO blood groups: I^AI^B = AB blood type)**

- Blood group O = ii; A = I^AI^A or I^Ai; B = I^BI^B or I^Bi; AB = I^AI^B (codominant)

#3 Inheritance of Two Genes

★★★ Very Imp.

★★★ **Dihybrid cross: RrYy × RrYy → Phenotypic 9:3:3:1; Genotypic 1:2:1:2:4:2:1:2:1**

★★★ **Epistasis: One gene masks/modifies expression of another non-allelic gene (gene interaction)**

- Recessive epistasis (9:3:4): e.g., Labrador coat colour (ee masks B locus)
- Dominant epistasis (12:3:1): dominant allele at one locus masks both alleles at another
- Complementary genes (9:7): both dominant alleles required for expression; absence of either = same phenotype

★★★ **Pleiotropy: One gene produces multiple phenotypic effects (e.g., PKU affects brain, skin, hair; sickle cell anaemia)**

#4 Chromosomal Theory, Linkage and Recombination

★★★ Very Imp.

★★★ **Sutton & Boveri (1902): Proposed chromosomes carry genes (Chromosomal Theory of Inheritance)**

★★★ **Morgan: Used *Drosophila melanogaster*; found genes on same chromosome — violated Mendel's 3rd law**

★★★ **Linkage: Genes on same chromosome inherited together; Complete (no crossing over) vs Incomplete (with crossing over)**

★★★ **Recombination frequency** $= \frac{\text{recombinant offspring}}{\text{total offspring}} \times 100\%$

★★ 1% recombination = 1 centimorgan (cM) = 1 map unit (m.u.); used to construct gene/chromosome maps

#5 Polygenic Inheritance and Pleiotropy

★★ Important

★★ Polygenic inheritance: Multiple genes (each with small additive effect) control one trait; shows **continuous variation**

- Example: Human skin colour controlled by 3 genes (A, B, C); ranges from very dark to very light; **bell-shaped distribution**

★★ Pleiotropy: Single gene → multiple phenotypic effects (PKU, sickle cell, Marfan syndrome)

#6 Sex Determination

★★★ Very Imp.

★★★ **Humans: XX = female; XY = male; sex determined by father (X-bearing or Y-bearing sperm)**

★★★ **Grasshopper: XX = female; XO = male (no Y chromosome; male is 2n-1)**

★★★ **Birds/Butterflies: ZW = female (heterogametic); ZZ = male — female determines sex of offspring**

★★★ **Honey bee: Fertilised egg (2n) → female worker/queen; Unfertilised egg (n) → male drone (arrhenotoky/parthenogenesis)**

- Barr body: Inactivated X chromosome in somatic cells of females (Lyon hypothesis); females have 1 Barr body per cell (2 X's)

#7 Mutation and Genetic Disorders

★★★ Very Imp.

★★★ **Mutation: Heritable change in DNA sequence; source of new alleles; raw material for evolution**

★★ Point mutation: Change in single base pair — e.g., Sickle cell anaemia: GAG → GUG (Glu → Val in β-globin)

★★★ **Chromosomal disorders (Aneuploidy):**

- **Down syndrome** (Trisomy 21, 47 chromosomes): Broad palm, furrowed tongue, intellectual disability, congenital heart defect; due to non-disjunction
- **Klinefelter's syndrome** (XXY, 47): Male phenotype; gynaecomastia; reduced facial/body hair; sterile
- **Turner's syndrome** (XO, 45): Female phenotype; short stature; webbed neck; underdeveloped gonads; sterile; horse-shoe kidney

★★★ **Mendelian genetic disorders:**

- Autosomal recessive: Sickle cell anaemia, PKU (phenylketonuria), Cystic fibrosis, Tay-Sachs, Alkaptonuria
- Autosomal dominant: Huntington's chorea (degeneration of neural tissue; appears after 30–40 yrs)
- X-linked recessive: Colour blindness (red-green; gene on Xq28), Haemophilia A (Factor VIII), Haemophilia B (Factor IX)

★★★ **Pedigree analysis: Square = male; Circle = female; Filled = affected; Horizontal line = mating; Vertical line = offspring; Used to determine inheritance pattern**

Chapter 5 | Molecular Basis of Inheritance

Day 3 · 7 Topics

#1 The DNA

★★★ Very Imp.

★★★ **Double helix model: Watson & Crick (1953); X-ray data from Rosalind Franklin (Photo 51)**

★★★ **Nucleotide = Phosphate + Deoxyribose sugar + Nitrogenous base; bases linked by glycosidic bond; nucleotides linked by 3'–5' phosphodiester bond**

★★★ **Purines: Adenine (A) + Guanine (G) (double ring); Pyrimidines: Thymine (T) + Cytosine (C) (single ring)**

★★★ **Chargaff's rules: A = T (2 H-bonds); G ≡ C (3 H-bonds); so A+G = T+C (purine = pyrimidine)**

★★★ **Double helix: antiparallel (5'→3' + 3'→5'); width = 2 nm; pitch = 3.4 nm; 10 bp per turn; right-handed B-form (usual)**

★★★ **Nucleosome: 146 bp DNA wound 1.65 turns around histone octamer (2×H2A + 2×H2B + 2×H3 + 2×H4); H1 = linker histone**

- 30 nm chromatin fibre → radial loops → scaffold → metaphase chromosome; level of compaction = 10,000×

#2 The Search for Genetic Material

★★★ Very Imp.

★★★ **Griffith (1928): Transformation experiment — heat-killed virulent S-strain + *live R-strain* → live S-strain appeared (mice died); showed a 'transforming principle'**

★★★ **Avery, MacLeod & McCarty (1944): Purified transforming principle = DNA (not RNA, protein, or polysaccharide)**

★★★ **Hershey & Chase (1952): T2 phage experiment: ^{32}P labels DNA; ^{35}S labels protein; DNA (not protein) entered bacteria → DNA is genetic material**

★★ RNA as genetic material: TMV (Tobacco Mosaic Virus) — RNA inoculated alone caused infection; also retroviruses (HIV)

- Properties of ideal genetic material: Self-replicate, stable, able to mutate, expressive (able to be translated)

#3 RNA World and Replication

★★★ Very Imp.

★★ RNA world hypothesis: RNA was first genetic + catalytic molecule (ribozyme); DNA evolved later for greater stability

★★★ **Semi-conservative replication proved by Meselson & Stahl (1958): E. coli grown in ^{15}N → shifted to ^{14}N → density gradient centrifugation confirmed hybrid then light DNA**

★★★ **Replication machinery: Helicase (unwinds) → SSBPs (stabilise) → Primase (RNA primer) → DNA Pol III (elongates 5'→3') → DNA Pol I (removes primer) → Ligase (joins)**

★★★ **Leading strand: synthesised continuously (same direction as replication fork); Lagging strand: synthesised in Okazaki fragments (discontinuous, joined by ligase)**

★★ Multiple origins of replication (ori) in eukaryotes allow faster replication of large genome

#4 Transcription

★★★ Very Imp.

★★★ **Central Dogma (Crick, 1958): DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ Protein (reverse: Reverse transcriptase in retroviruses)**

★★★ **Template strand (antisense, 3'→5'): read by RNA Pol; Coding strand (sense, 5'→3'): same sequence as mRNA (except T→U)**

★★★ **RNA Pol binds promoter (TATA box in eukaryotes) via transcription factors → unwinds DNA → synthesises mRNA 5'→3' → releases at terminator**

★★★ **Eukaryotic hnRNA processing (post-transcriptional modifications):**

- 5' capping: 7-methylguanosine added → protects from exonuclease; ribosome recognition
- 3' poly-A tail: 200+ adenine residues added → mRNA stability, export from nucleus
- Splicing: Introns removed; exons joined by spliceosome (snRNPs); produces mRNA ready for translation

★★ Monocistronic mRNA in eukaryotes (one ORF) vs polycistronic mRNA in prokaryotes (multiple ORFs, e.g., lac operon)

#5 Genetic Code

★★★ Very Imp.

★★★ **Codon: Triplet of mRNA nucleotides coding for one amino acid; $4^3 = 64$ codons for 20 amino acids**

★★★ **AUG: Start codon (methionine/fMet); also initiates translation; UAA, UAG, UGA: Stop/nonsense codons — signal termination**

★★★ **Properties of genetic code:**

- Universal: Same in almost all organisms (exceptions: mitochondria, Mycoplasma)
- Degenerate/Redundant: Most amino acids have multiple codons (e.g., Leu = 6 codons, Met = 1 codon)
- Non-ambiguous: Each codon specifies only ONE amino acid
- Non-overlapping, comma-less: Read continuously in triplets without gaps

★★ Wobble hypothesis (Crick): 3rd position of codon (wobble position) is flexible → explains degeneracy; one tRNA can recognise multiple codons

- Code cracked by Nirenberg & Matthaei (1961: poly-U → poly-Phe); completed by Khorana (synthetic codons)

#6 Translation and Regulation of Gene Expression

★★★ Very Imp.

★★★ **Translation: mRNA → polypeptide chain; requires mRNA, ribosomes, tRNAs, amino acids, ATP, GTP, factors**

★★★ **tRNA (adaptor molecule): Clover-leaf 2D / L-shaped 3D; Anticodon loop (recognises mRNA codon) + 3'-CCA-OH (amino acid attachment via aminoacyl-tRNA synthetase)**

★★★ **Ribosomes: 70S (prokaryotes: 50S + 30S); 80S (eukaryotes: 60S + 40S); three sites: A (aminoacyl), P (peptidyl), E (exit)**

★★★ **Translation steps:**

- **Initiation**: Small subunit binds mRNA at AUG; initiator tRNA (Met) occupies P site; large subunit joins
- **Elongation**: Aminoacyl-tRNA enters A site → peptidyl transferase forms peptide bond → translocation (ribosome moves 3 nt in 5'→3' direction)
- **Termination**: Stop codon at A site → release factor binds → polypeptide released; ribosome dissociates

★★★ **Lac Operon (Jacob & Monod, 1961 — Nobel Prize 1965): Inducible operon in *E. coli***

- Structural genes: *lacZ* (β-galactosidase — hydrolyses lactose), *lacY* (permease), *lacA* (transacetylase)
- No lactose: Repressor (from *lacI*) binds operator → RNA Pol blocked → genes **OFF**
- Lactose present: Allolactose (inducer) binds repressor → repressor can't bind operator → genes **ON**

★★★ **Trp operon: Repressible operon; tryptophan (co-repressor) + aporepressor → active repressor → genes OFF when Trp abundant**

#7 Human Genome Project & DNA Fingerprinting

★★★ Very Imp.

★★★ **HGP (1990–2003): International project to sequence entire human genome; Cost ~$3 billion**

★★★ **Key findings: ~3×10^9 bp per haploid set; ~20,000–25,000 protein-coding genes; only ~1.5% DNA codes for protein; 99.9% identical between any two humans**

★★ Approach: BAC/YAC vectors used as reference; Whole Genome Shotgun (WGS) sequencing; bioinformatics crucial

★★★ **DNA Fingerprinting (Alec Jeffreys, 1985): Based on VNTR (Variable Number Tandem Repeats) — unique per individual (except identical twins)**

★★★ **Steps: DNA extraction → RE digestion → gel electrophoresis → Southern blotting → hybridisation with radiolabelled probe → autoradiography → pattern (DNA profile)**

★★★ **Applications: Forensics (rape/murder), Paternity/maternity disputes, Identifying disaster victims, Population genetics, Evolutionary studies**

Chapter 6 | Evolution

Day 3 · 5 Topics

#1 Origin of Life and Evolution of Life Forms

★★★ Very Imp.

★★★ **Oparin (1923) & Haldane: Primordial (prebiotic) soup hypothesis — complex organics formed abiotically from inorganic molecules in reducing atmosphere**

★★★ **Primitive Earth atmosphere: CH_4, NH_3, H_2O, H_2 — NO free O_2 (reducing conditions)**

★★★ **Miller & Urey Experiment (1953): Simulated primitive atmosphere + electric sparks → amino acids formed in 15 days — proved abiogenesis**

★★ First self-replicating RNA → enclosed in lipid vesicles (proto-cells) → first life ~3.8 billion years ago (chemosynthetic prokaryotes)

- Coacervates (Oparin) and Microspheres (Fox): Proposed as first cell-like aggregates before true cells

#2 Evidences for Evolution

★★ Important

★★★ **Homologous organs: Same basic structure (same embryological origin), different functions → divergent evolution**

  - Example: Vertebrate forelimb — human arm, bat wing, whale flipper, horse foreleg (all have humerus, radius, ulna)

★★★ **Analogous organs: Different structure, similar function → convergent evolution (no common ancestry)**

  - Example: Bird wing & Butterfly wing; Shark fin & Dolphin flipper; Sweet potato (modified root) & Potato (modified stem)

★★ Vestigial organs: Reduced, non-functional remnants of ancestral structures (e.g., human appendix, coccyx, nictitating membrane, ear muscles)

- Biochemical evidence: Same genetic code, same ATP, same cytochrome-c across all life forms → common ancestry
- **Archaeopteryx**: Transitional fossil linking reptiles and birds; had feathers (bird) AND teeth, claws, bony tail (reptile)

#3 Adaptive Radiation, Biological Evolution, Mechanisms

★★★ Very Imp.

★★★ **Adaptive Radiation: Divergence from single common ancestor into multiple forms filling different ecological niches**

  - Darwin's finches (Galapagos Islands): 14 species from 1 ancestor; beak shapes differ by diet
  - Australian marsupials: Diverse from one ancestor; convergent with placental mammals on other continents

★★ Lamarck's theory: Use and disuse + inheritance of acquired characters (e.g., giraffe neck) — **DISPROVED**

★★★ **Darwin's Natural Selection: Variation in population → struggle for existence → survival & reproduction of fittest → inheritance of favourable traits**

★★★ **Mechanisms of evolutionary change:**

  - **Mutation**: Random heritable DNA changes; source of new alleles
  - **Natural selection**: Stabilising (mean favoured), Directional (one extreme), Disruptive (both extremes favoured)

- **Genetic drift**: Random allele frequency change in small populations; Founder effect; Bottleneck effect
- **Gene flow**: Movement of alleles between populations through migration

★★★ **Speciation: Allopatric (geographic isolation → reproductive isolation) vs Sympatric (same location, different niche/behaviour, e.g., polyploidy in plants)**

#4 Hardy-Weinberg Principle

★★★ Very Imp.

★★★ **In a large, randomly mating population: allele frequencies and genotype frequencies remain constant across generations (genetic equilibrium = no evolution)**

★★★ **Equation:** $p^2 + 2pq + q^2 = 1$; **and** $p + q = 1$
where p = frequency of dominant allele; q = frequency of recessive allele

★★★ **5 conditions for H-W equilibrium: (1) Large population (2) Random mating (3) No mutation (4) No natural selection (5) No gene flow**

★★ ANY violation of these 5 conditions = evolution is occurring in that population

- Application: Calculate carrier frequency; e.g., if $q^2 = 1/10000$ (affected), then q = 0.01; p = 0.99; carrier = 2pq ≈ 0.02 = 1 in 50

#5 Brief Account of Evolution & Origin of Man

★★★ Very Imp.

★★★ **Geological timeline:**

- 4.5 bya: Earth formed; 3.8 bya: First life (prokaryotes); 2.5 bya: O_2 accumulates (cyanobacteria)
- 500 mya: Invertebrates; 400 mya: Jawless fish; 350 mya: Land plants; 320 mya: Reptiles
- 200 mya: Mammals + birds diverge; 65 mya: K-Pg extinction (asteroid) → dinosaurs extinct → mammals diversify

★★★ **Human evolution:**

- 15 mya: Dryopithecus (ape-like) & Ramapithecus (more human-like)
- 3–4 mya: **Australopithecus** (walked upright; Africa; brain ~400 cc; used stone tools)
- 2 mya: **Homo habilis** (first genus *Homo*; brain ~700 cc; made tools; 'handy man')
- 1.5 mya: **Homo erectus** (brain ~900 cc; left Africa; spread to Asia & Europe; used fire; ate cooked meat)
- 1 mya: **Homo neanderthalensis** (brain ~1400 cc; used hides; buried dead; Neanderthals)
- ~75,000–10,000 ya: **Homo sapiens sapiens** (Cro-Magnon; modern humans; cave paintings; sophisticated tools; agriculture)

★★ Out of Africa hypothesis: Modern humans originated in Africa ~200,000 ya; migrated worldwide; replaced archaic humans

Chapter 7 | Human Health and Diseases

Day 4 · 5 Topics

#1 Common Diseases in Humans

★★★ Very Imp.

★★★ Typhoid fever: *Salmonella typhi* (bacteria); spread via contaminated food/water; Widal test for diagnosis; intestinal perforation if untreated; rose-colored spots on abdomen

★★★ Pneumonia: *Streptococcus pneumoniae*, *Haemophilus influenzae*; alveoli fill with fluid + mucus; lips/fingertips turn greyish-blue (cyanosis); spread via droplets

★★★ Malaria: *Plasmodium* (protozoan); vector = female *Anopheles* mosquito

- *P. falciparum*: Most deadly; malignant tertian malaria; can cause cerebral malaria, organ failure
- *P. vivax*: Benign tertian malaria; *P. malariae*: Quartan malaria; *P. ovale*: Least common
- Life cycle: Sporozoites (salivary gland) injected → liver cells (schizogony) → RBCs (merozoites) → rupture (fever/chills) → gametocytes in blood → mosquito (sexual cycle)

★★★ Amoebiasis (Amoebic dysentery): *Entamoeba histolytica*; contaminated food/water with cysts; blood and mucus in stool; cysts passed in faeces (infective)

★★★ Ascariasis: *Ascaris lumbricoides* (roundworm); contaminated food/water/soil; muscle pain, intestinal obstruction, can migrate to liver/lungs

★★★ Filariasis (Elephantiasis): *Wuchereria bancrofti*; vector = *Culex* mosquito; microfilariae in lymph vessels → blockage → gross swelling of limbs/scrotum

★★★ Ringworm: *Microsporum*, *Trichophyton* (fungi); contact transmission; dry, scaly, itchy circular lesions; spreads in warm/humid conditions

- Common cold: Rhinovirus (>100 serotypes); droplet + contact transmission; **antibiotics ineffective** (viral)

#2 Immunity

★★★ Very Imp.

★★★ Innate Immunity (Non-specific; present from birth; no memory):

- Physical barriers: Skin, mucus in respiratory/GI tract, wax in ear
- Physiological: Stomach HCl, tears (lysozyme), saliva (lysozyme), fever (inhibits pathogens)
- Cellular: Neutrophils, Monocytes, Macrophages, NK cells — phagocytose/lyse pathogens
- Cytokine barriers: Interferons (α, β, γ) released by virus-infected cells → protect neighboring cells

★★★ Acquired/Adaptive Immunity (Specific; develops after exposure; has memory):

★★★ Humoral Immunity (B-cell mediated): B-lymphocytes differentiate into plasma cells → secrete antibodies (immunoglobulins)

- Ig types: IgG (most abundant; crosses placenta), IgM (first in primary response; pentamer), IgA (secretions; colostrum), IgE (allergy; parasites), IgD (B-cell activation)
- **Primary response**: Slow, low titre, mainly IgM (first exposure to antigen)
- **Secondary/Anamnestic response**: Fast, high titre, mainly IgG (memory B-cells activated on re-exposure)

★★★ Cell-Mediated Immunity (CMI): T-lymphocytes; responsible for graft rejection, killing virus-infected cells

- T-helper ($CD4^+$): Coordinate immune response; activate B-cells + cytotoxic T-cells; major target of HIV
- T-cytotoxic ($CD8^+$): Kill infected cells directly via perforin + granzymes
- T-regulatory: Suppress immune response; prevent autoimmunity

★★★ **Active immunity: Body produces own antibodies (vaccination = deliberate active immunity; safe antigen/attenuated pathogen given)**

★★★ **Passive immunity: Pre-formed antibodies transferred — Maternal IgA (colostrum), Anti-tetanus serum, Anti-snake venom, Rabies immunoglobulin; immediate but short-lived**

★★ MHC (Major Histocompatibility Complex): HLA (Human Leukocyte Antigen) in humans; determines self vs non-self; responsible for graft rejection and autoimmunity

★★ Allergy: Exaggerated IgE-mediated response to harmless antigens (allergens); mast cells + basophils release **histamine + serotonin** → inflammation, bronchoconstriction

★★★ **Autoimmune diseases: Immune system attacks own cells: Rheumatoid arthritis (joints), Type-I diabetes (pancreatic β-cells), SLE, Multiple sclerosis**

#3 AIDS

★★★ Very Imp.

★★★ **Causative agent: HIV (Human Immunodeficiency Virus) — a Retrovirus (ssRNA genome; uses reverse transcriptase)**

★★★ **HIV specifically targets and destroys CD4$^+$ T-helper lymphocytes → immune system collapses → AIDS**

★★★ **Transmission routes: (1) Unprotected sexual contact (2) Contaminated blood/blood products/organ transplant (3) Shared needles (IV drug users) (4) Mother → child (placenta, during birth, breastfeeding)**

★★★ **NOT transmitted by: Hugging, handshake, sharing utensils or toilet, coughing/sneezing, mosquito bites**

★★ HIV replication: HIV enters CD4$^+$ cell → RT: RNA → dsDNA (provirus) → integrates into host chromosome → viral proteins produced → new virions bud off → cell dies

★★★ **Diagnosis: ELISA (screening test detects anti-HIV antibodies); confirmed by Western blot; CD4 count (normal = 600–1200 cells/mL; AIDS diagnosed when CD4 < 200)**

★★ Treatment: **ART (Antiretroviral Therapy / HAART)**: cocktail — NRTIs + NNRTIs + Protease inhibitors; controls viral load; no cure; **no vaccine yet**

> ***Note:*** *Window period: 8–10 weeks post-infection when ELISA can give false-negative (antibodies not yet at detectable levels). Patient is infectious during this period.*

#4 Cancer

★★★ Very Imp.

★★★ **Cancer: Uncontrolled cell proliferation; loss of contact inhibition; cells become invasive**

★★★ **Proto-oncogenes: Normal genes controlling cell division $\xrightarrow{\text{mutation/activation}}$ Oncogenes → uncontrolled growth**

★★★ **Tumour suppressor genes (p53, Rb): Normally prevent excess division; loss of function → cancer**

★★★ **Tumour types: Benign (localised, encapsulated, does NOT spread); Malignant (invasive; metastasises via blood/lymph)**

★★ Carcinogens: Tobacco (lung), UV radiation (skin/melanoma), X-rays; Viral: HPV → cervical cancer; HBV → liver cancer; EBV → Burkitt's lymphoma

★★★ **Detection: Biopsy + histopathology (gold standard), MRI, CT scan, X-ray; Tumour markers: PSA (prostate), CEA, AFP in blood**

★★★ **Treatment: Surgery + Radiotherapy (ionising radiation kills rapidly dividing cells)**

+ Chemotherapy (anti-mitotic drugs)

- α-interferon (biological response modifier): inhibits tumour growth; used clinically

#5 Drugs and Alcohol Abuse

★★★ Very Imp.

★★★ **Opioids: Morphine, Heroin (diacetylmorphine); extracted from latex of *Papaver somniferum* (opium poppy); CNS depressant; highly addictive; sedation + euphoria + pain relief**

★★★ **Cannabinoids: THC from *Cannabis sativa* (leaves, resin); affect cardiovascular system; forms: Marijuana/Ganja (dried leaves), Hashish (resin), Charas**

★★★ **Cocaine (Coca alkaloid): From *Erythroxylum coca*; powerful CNS stimulant; blocks dopamine reuptake → euphoria; snorted/injected; highly addictive**

- Other drugs: Barbiturates, Benzodiazepines, Amphetamines (speed), LSD (hallucinogen)

★★ Addiction: Psychological & physical dependence; tolerance develops (need more for same effect); withdrawal causes physical illness

★★★ **Alcohol effects: Liver cirrhosis → liver failure; peripheral neuritis; Wernicke-Korsakoff syndrome (brain); Foetal Alcohol Syndrome; impaired judgment/reflexes**

- Anabolic steroids (athletes): Liver damage, acne, mood swings ('roid rage'), testicular atrophy, virilisation in females, stunted growth in teenagers

> ***Note:*** *DOPE test (urinalysis/blood test) detects banned substances in competitive sports. Anabolic steroids are banned by WADA.*

Chapter 8 | Microbes in Human Welfare

Day 4 · 6 Topics

#1 Microbes in Household Products

★★★ Very Imp.

★★★ **Curd/Yogurt: *Lactobacillus* / *Lactococcus lactis* → lactic acid fermentation → curdling of milk; increases Vit. B12; small amount of curd added as 'starter' (inoculum)**

★★★ **Bread: *Saccharomyces cerevisiae* (baker's yeast) → ferments sugars → CO_2 bubbles → bread rises (leavening); dead yeast contributes to bread flavour**

★★ Idli and Dosa: Batter of rice + black gram fermented by *Leuconostoc mesenteroides* → lactic acid + CO_2

- Cheese: *Lactococcus* (cottage); *Propionibacterium shermanii* (Swiss cheese holes); *Penicillium roquefortii* (Roquefort blue cheese)
- Toddy: Natural fermentation of palm tree sap (*Caryota urens*) by naturally occurring microbes

#2 Microbes in Industrial Products

★★★ Very Imp.

★★★ **Fermented beverages (*Saccharomyces cerevisiae* — anaerobic fermentation):**

- Wine & Beer: WITHOUT distillation (wine <13% ethanol; beer <8% + CO_2)
- Whisky, Brandy, Rum: WITH distillation; brandy from fruit juice; rum from molasses

★★★ **Antibiotics (MOST IMPORTANT):**

- **Penicillin**: *Penicillium notatum* (Fleming 1928; Nobel 1945; Chain & Florey developed; first antibiotic; kills gram-positive bacteria by inhibiting cell wall synthesis)
- Streptomycin: *Streptomyces griseus*; Tetracycline: *Streptomyces*; Erythromycin: *Streptomyces erythreus*

★★★ **Organic acids:**

- Citric acid: *Aspergillus niger*; Acetic acid: *Acetobacter aceti*; Butyric acid: *Clostridium butylicum*; Lactic acid: *Lactobacillus*

★★★ **Industrial enzymes:**

- Lipases: In detergents (remove oil/grease stains at low temperature)
- Proteases: Meat tenderisers, cheese production, detergents
- Pectinases + Amylases: Clarify bottled fruit juices and wines
- **Streptokinase**: From *Streptococcus*; 'clot buster' — dissolves fibrin clots in heart attacks (thrombolytic agent)
- Cyclosporin A: *Trichoderma polysporum* — immunosuppressant in organ transplantation

★★★ **Statins: *Monascus purpureus* (red yeast rice); e.g., Lovastatin; competitively inhibit HMG-CoA reductase (key enzyme in cholesterol synthesis) → lower blood cholesterol**

#3 Microbes in Sewage Treatment

★★★ Very Imp.

★★★ **Primary treatment (Physical): Bar screens → grit chamber → sedimentation tank (primary sludge settles; effluent decanted); removes suspended solids (35% BOD reduction)**

★★★ **Secondary treatment (Biological): Primary effluent + activated sludge + air pumped into aeration tanks → heterotrophic microbes form flocs (masses) → digest organic matter → BOD reduced by >90%**

★★★ **BOD (Biochemical Oxygen Demand): O_2 required by microbes to oxidise all organic**

matter in 1L water; lower BOD = cleaner water

★★ Activated sludge: Dense microbial flocs; part recycled to aeration tank; remainder sent to **anaerobic sludge digester**

★★★ **Anaerobic sludge digester: Methanogens break down organic matter → biogas (CH_4 + CO_2 + H_2S) + digested sludge (used as manure)**

#4 Microbes in Production of Biogas

★★ Important

★★ Biogas: Mainly **CH_4** (methane) + CO_2 + H_2S; produced by **Methanogens** (archaebacteria: *Methanobacterium*) in strictly anaerobic conditions

- Biogas plant: Concrete tank → slurry inlet (cattle dung + water 1:1) → floating gas holder (collects gas) → gas outlet → spent slurry outlet
- KVIC (Khadi and Village Industries Commission): Designed and popularised biogas plants in rural India
- Spent slurry: Excellent organic manure; rich in N and P; safe to handle

#5 Microbes as Biocontrol Agents

★★ Important

★★ *Bacillus thuringiensis* (Bt): Forms protein crystals (Cry proteins); ingested by insect larvae → alkaline gut activates toxin → binds gut epithelium → cell lysis → death; **harmless to mammals**

  - Different Bt strains: Bt var. *kurstaki* (Lepidoptera); Bt var. *israelensis* (Diptera/mosquitoes); Bt var. *tenebrionis* (beetles)

- *Trichoderma*: Free-living soil fungus; biological control of plant pathogens (*Pythium*, *Rhizoctonia*, *Fusarium*)
- Baculoviruses (Nucleopolyhedrovirus / NPV): Species-specific insect control; safe for non-target organisms, birds, humans

> ***Note:*** *IPM (Integrated Pest Management): Combines biocontrol + resistant varieties + minimum chemical use = sustainable agriculture.*

#6 Microbes as Biofertilisers

★★★ Very Imp.

★★★ **Nitrogen-fixing microbes (fix atmospheric N_2 → NH_3):**

  - Free-living: *Azotobacter* (aerobic), *Clostridium* (anaerobic), Cyanobacteria (*Anabaena, Nostoc, Oscillatoria*)
  - Symbiotic: ***Rhizobium*** (in root nodules of legumes — most important); *Frankia* (non-legume trees like *Alnus*); *Anabaena* in *Azolla* (water fern, used in paddy fields)

★★★ **Mycorrhiza: Symbiotic fungus–plant root association; fungal hyphae enormously increase surface area for absorption**

  - Ectomycorrhiza: Hyphae surround root (e.g., *Pinus*); Endomycorrhiza / **VAM**: *Glomus* penetrates root cells → greatly increases phosphate + water absorption; drought/pathogen resistance

- PSB (Phosphate Solubilising Bacteria): *Bacillus, Pseudomonas* — convert insoluble phosphates to soluble form

> ***Note:*** *Biofertilisers are eco-friendly, reduce input costs, improve soil health, and do not cause pollution unlike chemical fertilisers.*

Chapter 9 | Biotechnology: Principles and Processes

Day 5 · 3 Topics

#1 Principles of Biotechnology

★★★ Very Imp.

★★★ **Modern Biotechnology rests on two core techniques: (1) Genetic engineering (rDNA technology); (2) Aseptic maintenance of cell/tissue cultures**

★★★ **Palindromic sequence: DNA sequence that reads identically on both strands in 5'→3' direction; recognition sites for restriction enzymes**

- Example: 5'-**G A A T T C**-3' / 3'-**C T T A A G**-5' (EcoRI site)

★★★ **Restriction Endonucleases ('molecular scissors'): Cut dsDNA at specific recognition sequences (usually 4–8 bp palindromes)**

- **EcoRI**: Cuts G↓AATTC → produces 4-nt 5' sticky (cohesive) ends; sticky ends allow easy ligation by hydrogen bonding
- Blunt-end cutters: e.g., SmaI, HaeIII; cut in middle of recognition sequence
- Naming: **E**co = organism (*E. coli*), **R** = strain (RY13), **I** = order of discovery

★★ Gel Electrophoresis: Agarose gel; DNA (negatively charged) migrates toward anode (+); smaller fragments travel farther; stained with **ethidium bromide** (EtBr) → orange bands under UV

#2 Tools of Recombinant DNA Technology

★★★ Very Imp.

★★★ **4 essential tools: Restriction enzymes + DNA Ligase + Cloning vector + Suitable host organism**

★★★ **Cloning vectors must have: (1) ori (origin of replication); (2) Selectable marker; (3) Cloning sites (MCS)**

★★★ **Types of vectors:**

- Plasmids: Circular, self-replicating; **pBR322**: first artificial recombinant plasmid; has amp^R, tet^R, *ori*, MCS sites
- Bacteriophage vectors: Lambda phage (λgt10/gt11); M13; useful for large inserts
- BAC (Bacterial Artificial Chromosome): 100–300 kb inserts; YAC (Yeast Artificial Chromosome): up to 1000 kb

★★★ **Insertional inactivation (selecting recombinants): Insert foreign gene into tet^R of pBR322 → recombinant cells grow on amp plate but NOT on tet plate (tet^R disrupted)**

★★★ **Blue-white screening: Insert in *lacZ* gene (pUC vectors) → recombinants = white (no β-galactosidase); non-recombinants = blue (active β-galactosidase makes blue product from X-gal)**

★★ Competent cells: $CaCl_2$ treatment + heat shock (42°C, 90 sec) → cells become permeable to DNA; Electroporation: electric pulse used for plant cells; Biolistics: DNA-coated gold particles fired into cells

#3 Processes of Recombinant DNA Technology

★★★ Very Imp.

★★★ **Complete process: (1) DNA isolation → (2) RE digestion → (3) Gel electrophoresis (isolate band) → (4) Ligation into vector → (5) Transformation into host → (6) Selection/screening → (7) Expression → (8) Downstream processing**

★★★ **PCR (Polymerase Chain Reaction, Kary Mullis 1983, Nobel 1993): In vitro amplification of specific DNA sequence**

- Components: Template DNA, Forward + Reverse primers (specific to target), **Taq polymerase** (from *Thermus aquaticus* — thermostable to 95°C), dNTPs, $MgCl_2$ buffer
- Cycle: **Denaturation** (94°C, 1 min) → **Annealing** (50–65°C, 30–60 sec) → **Extension** (72°C, 1 min); each cycle doubles copies; 30 cycles ≈ 10^9 copies

★★★ **Bioreactor: Large vessel (100–1000 L+) for industrial-scale cell culture/fermentation**

- Stirred-tank bioreactor: Most common; sparger (air supply), impeller (mixing), pH + temp + O_2 sensors; water jacket for temperature control

★★ Downstream processing: After fermentation — extraction → purification (chromatography, filtration) → formulation → clinical trials → market

Chapter 10 | Biotechnology and Its Applications

Day 5 · 4 Topics

#1 Biotechnological Applications in Agriculture

★★★ Very Imp.

★★★ Bt crops: *Bacillus thuringiensis cry* protein genes → introduced into crop plant genome → plant produces insecticidal Cry toxins itself

- Bt cotton (approved India 2002): *cry1Ac* + *cry2Ab* genes → protection against cotton bollworm (*Heliothis/Helicoverpa*)
- Bt corn: *cry1Ab* gene → corn borer (*Ostrinia*)

★★★ Mechanism of Bt toxin: Cry protoxin (inactive crystals) → ingested by insect → alkaline midgut dissolves crystals → activated toxin binds cadherin receptor on gut epithelium → osmotic lysis → insect dies; safe for mammals (acidic stomach, no receptor)

★★★ RNAi (RNA interference): dsRNA introduced into plant → RISC (RNA-induced silencing complex) degrades complementary mRNA → gene silenced

- Example: Tobacco against root nematode (*Meloidogyne incognita*) — nematode-specific dsRNA introduced → nematode gene silenced in plant → parasite dies

★★★ Golden Rice: Transgenic rice with β-carotene biosynthesis genes from *Narcissus* + *Erwinia* (bacteria) expressed in endosperm → rice is yellow-golden; addresses Vitamin A deficiency in Southeast Asia

★★ GEAC (Genetic Engineering Appraisal Committee, India): Regulatory body approving safety and release of GMOs in India

- Herbicide-tolerant crops (Roundup Ready soybean): Modified EPSPS gene → tolerates glyphosate herbicide application

#2 Biotechnological Applications in Medicine

★★★ Very Imp.

★★★ Recombinant Human Insulin (Humulin, Eli Lilly, 1982): First recombinant pharmaceutical approved by FDA

- Normal insulin: A chain (21 aa) + B chain (30 aa) linked by 2 disulfide bonds; C-peptide (31 aa) connects them in proinsulin
- Production: A & B chains produced **separately** in *E. coli* (using separate recombinant plasmids) → extracted → disulfide bonds formed in vitro → functional insulin

★★★ Gene therapy: Introduction of functional gene copy to permanently correct genetic defect

- ADA (Adenosine Deaminase) deficiency (SCID): First successful gene therapy (1990); functional ADA gene introduced into patient's T-lymphocytes via retroviral vector; permanent cure requires introduction into haematopoietic stem cells

★★★ Molecular Diagnostics:

- **ELISA**: Enzyme-Linked ImmunoSorbent Assay; detects antibodies OR antigens; used for HIV, hepatitis, pregnancy test, food allergens; highly sensitive
- **PCR-based diagnosis**: Detects even a single copy of pathogen DNA/RNA; used for HIV (before antibodies form), TB, cancer gene mutations, prenatal diagnosis
- Hybridisation with radiolabelled/fluorescent DNA probes: Detects specific pathogen nucleic acid sequences

- Recombinant vaccines: Hepatitis-B vaccine (HBsAg from yeast); HPV vaccine (Gardasil); no risk of infection from vaccine

- Other products: Human Growth Hormone (somatotropin), Clotting factors VIII & IX (haemophilia), Erythropoietin (EPO for anaemia), Interferon-alpha (cancer/hepatitis)

#3 Transgenic Animals

★★ Important

★★ Transgenic animals: Animals with foreign gene (transgene) stably integrated into genome in all cells including germ cells

- Production: Micro-injection into fertilised egg (most common); retroviral vectors; embryonic stem (ES) cells

★★ **Rosie the cow**: Produced human α-1-antitrypsin (for emphysema treatment) in her milk at 2.4 g/L

- Disease models: Transgenic mice for Alzheimer's (APP gene), cancer, cystic fibrosis, AIDS; used to test new treatments
- Vaccine safety testing (polio vaccine); chemical safety testing (carcinogenicity); nutritional studies

★★ **Dolly the sheep** (Ian Wilmut, Edinburgh, 1996): First mammal cloned from adult somatic cell using SCNT (Somatic Cell Nuclear Transfer); confirmed **totipotency** of somatic cells

#4 Ethical Issues

★★ Important

★★ Biopiracy: Unauthorised, unfair commercial exploitation of biological resources or traditional knowledge without compensation

- Examples: Basmati rice (USA patent challenged by India); Turmeric wound-healing property; Neem (antifungal) — all challenged and revoked

★★★ **CBD (Convention on Biological Diversity, 1992): Nations have sovereign rights over their biological resources; requires prior informed consent (PIC) + benefit sharing**

- IPR (Intellectual Property Rights): TRIPS (Trade-Related Intellectual Property Rights) agreement; biopatents protect novel biological inventions

★★ India: Biological Diversity Act 2002; National Biodiversity Authority; amended patent laws to prevent biopiracy

★★ Ethical concerns on GMOs: Potential unknown ecological impact, horizontal gene transfer to wild relatives, reduced genetic diversity, corporate monopoly on seeds (Terminator gene)

> ***Note:*** *International bodies (WHO, FAO, CBD) collaborate on the Cartagena Protocol on Biosafety to regulate transboundary movement of GMOs.*

Chapter 11 | Organisms and Populations

Day 6 · 3 Topics

#1 Population Attributes

★★★ Very Imp.

★★★ **Population: Group of individuals of same species sharing same geographic area at same time**

★★★ **Key population attributes: Natality (birth rate), Mortality (death rate), Age structure, Sex ratio, Population density**

★★★ **Population density: Number of individuals per unit area or volume; also measured as biomass, percentage cover (plants), pheromone traps (insects)**

★★ Age pyramid: **Expanding** (broad base — growing population); **Stable** (equal distribution — stable); **Declining** (narrow base — decreasing)

#2 Population Growth

★★★ Very Imp.

★★★ **Exponential growth (J-shaped curve):** $\frac{dN}{dt} = rN$ **where r = intrinsic rate of natural increase; occurs when resources are unlimited**

- Integral form: $N_t = N_0 \cdot e^{rt}$; population doubles in fixed time ($t_d = \frac{0.693}{r}$)

★★★ **Logistic/Verhulst-Pearl growth (S-shaped / sigmoid curve):** $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

- K = carrying capacity (maximum population the environment can sustainably support); growth slows as N approaches K
- Inflection point (maximum growth rate) at **N = K/2**

★★★ **r (intrinsic rate of natural increase) = birth rate (b) − death rate (d)**

★★ K-selected species: Low r, large body, long lifespan, few offspring, parental care (elephants, whales, humans)

★★ r-selected species: High r, small body, short lifespan, many offspring, no parental care (insects, bacteria, rodents)

#3 Life History Variation and Population Interactions

★★★ Very Imp.

★★★ **Population interactions (+ = benefit; − = harm; 0 = unaffected):**

Interaction	Species A	Species B	Example
Mutualism	+	+	Rhizobium + legume; Mycorrhiza
Competition	−	−	Two species for same food
Predation	+	−	Lion + deer; plant + herbivore
Parasitism	+	−	Plasmodium + human
Commensalism	+	0	Orchid on tree; Cattle egret
Amensalism	−	0	Penicillium vs bacteria

★★★ **Gause's Competitive Exclusion Principle: Two species competing for identical resources cannot coexist indefinitely; one will outcompete and eliminate the other**

★★★ **Predation: Regulates prey population; drives coevolution; plants evolved defences (thorns: Acacia; latex: Calotropis; chemicals: morphine, caffeine, quinine, tannin)**

★★★ **Parasitism: Ectoparasites (external: lice, ticks, mites, leech); Endoparasites (internal:**

tapeworm, Ascaris, Plasmodium); Brood parasitism (Cuckoo + Crow)

★★★ **Commensalism: Orchid on mango (epiphyte; uses tree for support, not nutrition); Cattle egret + cattle (eats insects disturbed by grazing); Clownfish in sea anemone**

★★ Batesian mimicry: Harmless species mimics harmful/distasteful species for protection (e.g., Viceroy butterfly mimics Monarch)

Chapter 12 | Ecosystem

Day 6 · 4 Topics

#1 Ecosystem — Structure and Function

★★★ Very Imp.

★★★ **Ecosystem: Functional unit = biotic community + abiotic environment interacting as a system; concept by A.G. Tansley (1935)**

★★★ **Components: Abiotic (sunlight, temperature, precipitation, soil, water) + Biotic (Producers, Consumers, Decomposers)**

★★★ **Trophic levels: T1 = Producers (autotrophs/plants); T2 = Primary consumers (herbivores); T3 = Secondary consumers; T4 = Tertiary consumers**

- Grazing food chain (GFC): Starts from living green plants; Detritus food chain (DFC): Starts from dead organic matter
- Food web: Multiple interconnected food chains; more complex food web = more stable ecosystem

★★ Standing crop: Amount of living biomass at each trophic level at any given time

#2 Productivity

★★★ Very Imp.

★★★ **GPP (Gross Primary Productivity): Total organic matter fixed by autotrophs via photosynthesis per unit area per unit time**

★★★ **NPP (Net Primary Productivity) = GPP − Respiration (R); organic matter available to consumers**

★★★ **Secondary productivity: Rate of formation of new organic matter by consumers; energy assimilated minus respiration**

★★ Highest NPP: Tropical rainforests + estuaries + coral reefs; Lowest NPP: Open ocean + desert

- Units: g dry weight m^{-2} yr^{-1} OR kcal m^{-2} yr^{-1}

#3 Decomposition

★★★ Very Imp.

★★★ **Decomposition: Breakdown of complex organic matter (detritus) → inorganic substances; carried out by decomposers (bacteria + fungi)**

★★★ **Steps of decomposition:**

- **Fragmentation**: Detritivores (earthworms, millipedes) physically break detritus into smaller particles
- **Leaching**: Water-soluble inorganic nutrients leach down into soil horizon
- **Catabolism**: Bacterial + fungal enzymes degrade polymers → simpler organic monomers
- **Humification**: Accumulation of **humus** (dark, amorphous, colloidal, resistant organic matter; improves soil structure)
- **Mineralisation**: Humus further decomposed → inorganic minerals (N, P, K, Ca) released back to soil

★★ Decomposition slower when: Low temperature (tundra), waterlogged/anaerobic conditions (peat bogs), high C:N ratio of detritus

#4 Energy Flow

★★★ Very Imp.

★★★ **10% Energy Law (Lindeman, 1942): Only ~10% of energy at one trophic level is transferred to the next; rest is lost as heat + respiration**

- Implication: Shorter food chain = more energy available; explains why large carnivores are rare

★★★ Ecological pyramids: Pyramid of Number, Pyramid of Biomass, Pyramid of Energy

- Pyramid of Number: Can be **inverted** (1 tree supports many insects supports few birds)
- Pyramid of Biomass: Can be **inverted** in aquatic ecosystems (phytoplankton biomass < zooplankton at any instant due to fast turnover)
- Pyramid of Energy: **ALWAYS UPRIGHT** (energy always decreases at each trophic level; most fundamental pyramid)

★★★ Biogeochemical cycles (nutrient cycling):

- **Carbon cycle**: Photosynthesis (CO_2 fixed) ↔ Respiration + Decomposition + Combustion (CO_2 released); reservoir = atmosphere (CO_2) + ocean + fossil fuels
- **Nitrogen cycle**: N-fixation ($N_2 \rightarrow NH_3$) → Nitrification ($NH_3 \rightarrow NO_3^-$) → Assimilation (plants absorb) → Ammonification (detritus → NH_3) → Denitrification ($NO_3^- \rightarrow N_2$)

> ***Note:*** *Ecosystem services (e.g., carbon sequestration, pollination, water purification, flood control) are worth an estimated $33 trillion/year globally.*

Chapter 13 | Biodiversity and Conservation

Day 6 · 6 Topics

#1 Biodiversity

★★★ Very Imp.

★★★ **Three levels: (1) Genetic diversity (allelic variation within species); (2) Species diversity (species richness in an area); (3) Ecosystem diversity (variety of ecosystems/biomes)**

★★★ **India: Mega-diverse country; 8.1% of global species richness on only 2.4% of Earth's land; ∼45,000 plant + ∼90,000 animal species**

- India has >50,000 strains of rice and 1,000 varieties of mango (genetic diversity example)

#2 How Many Species on Earth and in India

★★★ Very Imp.

★★★ **Estimated ∼8.7 million species on Earth; only ∼1.5 million formally described and named; majority yet undiscovered (especially tropical + marine)**

★★ More than 70% of all species found in tropics; invertebrates dominate total species count; fungi and microbes add millions more

- India has 4 biodiversity hotspots + very high endemism (species found only in India)

#3 Patterns of Biodiversity and Importance of Species Diversity

★★★ Very Imp.

★★★ **Latitudinal gradient: Species richness increases from poles to tropics; tropics have highest diversity**

- Reasons: Greater solar energy input; stable climate over millions of years; more complex niches; more time for speciation; higher productivity

★★★ **Species-Area relationship (Arrhenius equation):** $\log S = \log C + Z \log A$

- S = species richness; A = area; Z = slope (regression coefficient); C = y-intercept
- Z = 0.1–0.2 for small areas within a region; Z = 0.6–1.2 for very large continental areas

★★★ **Importance of species diversity: Greater diversity → more stable, productive, and resilient ecosystem**

★★ **Rivet popper hypothesis** (Paul Ehrlich): Ecosystem = airplane; each species = rivet; removing rivets (species) weakens the plane; enough rivets removed → sudden catastrophic collapse

#4 Loss of Biodiversity

★★★ Very Imp.

★★★ **Current extinction rate: 100–1000× greater than background (pre-human) extinction rate**

★★★ **The 'Evil Quartet' — four major causes of biodiversity loss:**

- 1. **Habitat loss and fragmentation**: MOST IMPORTANT; deforestation, urbanisation, agriculture; especially catastrophic in tropics; leads to island-like forest patches
- 2. **Over-exploitation**: Hunting, poaching, overfishing beyond sustainable levels; extinctions: Steller's sea cow (1768), Passenger pigeon (1914), Dodo, Great Auk
- 3. **Alien (exotic) species invasions**: Introduced species outcompete natives; Nile perch in Lake Victoria (→ eliminated >200 endemic cichlid fish); Parthenium, Lantana, *Eichhornia* (water hyacinth) in India
- 4. **Co-extinctions**: Obligate mutualists go extinct together (host plant extinct → specialist pollinator/herbivore extinct)

★★ Island biogeography: Smaller area → fewer species can persist; fragmentation creates islands of habitat

#5 Biodiversity Conservation — Why Should We Conserve?

★★ Important

★★ Narrowly utilitarian: Direct economic value — food, ∼25% of all pharmaceutical drugs from plants, fibre, fuel, timber

★★ Broadly utilitarian: Ecosystem services — pollination (food security), O_2 production, climate regulation, water purification, soil formation, flood control; worth ∼$33 trillion/year

- Ethical: Intrinsic value — every species has a right to exist, regardless of utility to humans
- Bioprospecting: Searching wild species for potential new medicines, industrial chemicals

#6 How Do We Conserve Biodiversity?

★★★ Very Imp.

★★★ **In-situ conservation (on-site, within natural habitat): Most effective approach**

- **Biodiversity hotspots**: 34 globally; defined by high endemism (>0.5% global plant endemics) + high threat (>70% habitat lost); **India has 4**: Western Ghats + Sri Lanka; Indo-Burma; Eastern Himalayas; Sundaland
- Biosphere Reserves (18 in India): Core zone (strictly protected) + Buffer zone (research/tourism) + Transition zone (human use)
- National Parks (106 in India): Strictly protected; no human activity; e.g., Jim Corbett, Kaziranga, Sundarbans
- Wildlife Sanctuaries (551 in India): Limited human use; e.g., Bharatpur, Periyar, Chinnar
- Sacred groves: Forest patches protected by local communities on religious grounds; found across India (Meghalaya, Rajasthan, etc.)

★★★ **Ex-situ conservation (off-site, outside natural habitat):**

- Zoological parks, Botanical gardens, Wildlife safari parks
- **Seed banks**: Long-term storage at −196°C (liquid N_2); Svalbard Global Seed Vault (Norway) = world's largest
- Gene banks, Gamete cryopreservation, Tissue culture, In vitro fertilisation

★★★ **IUCN Red List categories: EX (Extinct) → EW (Extinct in Wild) → CR (Critically Endangered) → EN (Endangered) → VU (Vulnerable) → NT (Near Threatened) → LC (Least Concern)**

★★★ **International conventions:**

- **CBD** (Convention on Biological Diversity, Rio 1992): India signatory; sovereign rights over biodiversity; benefit sharing
- **CITES**: Convention on International Trade in Endangered Species; regulates/bans wildlife trade
- **Ramsar Convention**: Protection of wetland habitats globally
- **World Heritage Convention** (UNESCO): Protects natural + cultural sites of outstanding universal value

— END OF REVISION GUIDE —

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